Sepideh Sepehri

Email: sepidehsepehri75@gmail.com Google Scholar: Sepideh Sepehri Github: Sepideh Sepehri

Education

MSc in Astrophysics & Cosmology *Oct 2022 – July 2025*
University of Padova, Italy
Supervisor: Prof. Giulia Rodighiero
Evolution of Galaxies and AGN Group, University of Padova, Italy
Thesis title: The Role of Galaxy Mergers in the High Redshift Universe with Deep JWST Extragalactic Fields
Analyzed ∼500,000 galaxies from ASTRODEEP–JWST to measure merger pair fractions up to $z \sim 10$. Developed Python pipeline using `Astropy, NumPy, SciPy` for catalog merging, S/N selection, color classification, and companion search (5–50 kpc). Applied completeness corrections, statistical uncertainties, and systematic-error assessments. Results suggest a possible enhancement in major merger pair fractions at $z \sim 8.5$, potentially linked to early environmental assembly or redshift uncertainties. Gained experience with wide-field extragalactic imaging, photometric catalogs, and cosmology-oriented data analysis.

BSc in Physics *Sept 2015 – Sept 2020*
Tehran Polytechnic (Amirkabir University of Technology), Iran

Research Projects

Machine Learning for Astronomical Data *Aug 2025 – Current*
Independent Research
Exploring ML applications in galaxy morphology and merger identification inspired by recent studies. Implemented basic regression, classification, and clustering models using `ckdTree`, and `Scikit-learn` and evaluated their performance on small test datasets.

Gravitational Waves: Boltzmann & Observational Approaches *Mar 2024 – Jul 2024*
Theoretical Cosmology Course, University of Padova, Italy
Supervisor: Prof. Sabino Matarrese
Reviewed gravitational wave generation using Boltzmann equations. Assessed signal detectability with LISA and the Einstein Telescope. Linked early-universe GWs to inflation and structure formation.

Photometric Transit Analysis of WASP-12b *Oct 2023 – Jan 2024*
Astrophysics Laboratory II, University of Padova, Italy
Supervisor: Prof. Luca Malavolta
Performed light-curve modeling of the exoplanet WASP-12b using TESS and TASTE photometry. Applied Python with the `batman` and `emcee` libraries for transit fitting and MCMC parameter estimation. Derived orbital and physical parameters consistent with literature and contributed to the final scientific report.

Publications

- **In Preparation:** "Quantifying the Impact of Mergers on Dusty Star Formation with JWST".
  - Sepehri, S., Gandolfi, G., Rodighiero, G., Bisigello, L., Girardi, G., Grazian, A., & collaborators.
- Current Focus: Integrating comprehensive spectroscopic redshift compilations (COSMOS spec-z) with JWST/ASTRODEEP photometric catalogs to investigate $z \sim 8.5$ pair-fraction anomalies in the PRIMER-COSMOS field and address photometric-redshift uncertainties.

Conferences & Seminars

"Quantifying Galaxy Mergers' Impact on Dusty SF with JWST and Euclid" *July 2025*
– European Astronomical Society Annual Meeting 2025, Cork, Ireland
Presented by: Dr. Giovanni Gandolfi. Results included from my M.Sc. thesis on JWST-based merger pair statistics.

Scientific Experiences

CubeSats: Fly Your Satellite! (ESA Program) *Feb 2023 – Sept 2023*
Ground Station Team, University of Padova, Italy
Worked on ground station operations for ESA-supported CubeSat. Studied satellite tracking, communication systems, and signal decoding. Gained experience with control software, link budgets, and VHF/UHF systems.

Observing Nights at Asiago Observatory *Feb 2023 & Jan 2024*
Astrophysics Laboratory I & II, Asiago Observatory, Italy
Participated in MSc observing sessions. Experience gain in photometry, spectroscopy, telescope control.

Certifications & Workshops

Machine Learning Specialization *Nov 2025*
Stanford University / Coursera
JWST Summer School: High Redshift Transients with JWST *Aug 2025*
Space Telescope Science Institute (STScI), Baltimore, USA
Statistical Analysis of Cosmic Fields *Sep – Nov 2021*
K. N. Toosi University of Technology, Tehran, Iran
Advanced Cosmology Course *Feb – July 2020*
Institute for Research in Fundamental Sciences (IPM), Tehran, Iran
Summer School: Introduction to Cosmology *June – Sept 2019*
Sharif University of Technology, Tehran, Iran

Scholarships & Awards

Veneto Regional Government Scholarship (ESU) *2022–2025*
Awarded a merit-based scholarship for international students for MSc studies.

Outreach Activities

Astronomy Outreach & Editorial Work *2018–2019*
Tehran Polytechnic Astronomy Center, Iran
Translated and edited astronomy content for the University Astronomy Center and contributed to the preparation of the Persian Astronomy Calendar (2018–2019).

Other Relevant Skills

Programming & Data Analysis:
(Advanced): Python: `Astropy, NumPy, SciPy, Pandas, Matplotlib, astroML`; LaTeX, GitHub.
(Basic–Intermediate): Machine learning: `scikit-learn, KDTree`; Bash / Linux shell;
(Basic): C/C++, R, SQL.
Astronomical Software & Tools:
TOPCAT; SAOImage DS9; SpecPro; SExtractor.
Languages: Persian (Native), English (C1), Italian (A2), German (A1)

References

- Giulia Rodighiero (Master Thesis Supervisor, UniPD, INAFPadua) — giulia.rodighiero@unipd.it
- Giovanni Gandolfi (Master Thesis Co-Supervisor, INAF, Rome) — giovanni.gandolfi@unipd.it